Milinda Fernando

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Education

2021 - Ph.D., Scientific Computing, The University of Utah, UT.
- **Supervisor**: Dr. Hari Sundar
- **Dissertation**: *"Scalable spacetime adaptivity for computational general relativity"*

2015 - B.Sc., Computer Science and Engineering, University of Moratuwa, Sri Lanka.
- Dissertation: *"Auto-tuning the Java virtual machine using OpenTuner framework"*

Work Experience

- **Sep. 2023 - Present** : **Research Associate**, Oden Institute, The University of Texas at Austin, TX.
- Jun. 2021 - Sep. 2023 : **Peter O'Donnell, Jr. Postdoctoral Fellow**, Oden Institute, The University of Texas at Austin, TX.

  Supervisor: Dr. George Biros

  Research: *During my time at the Oden Institute, my research focuses on developing a deterministic Eulerian Boltzmann transport equation (BTE) solver for electron transport in low-temperature weakly-ionized plasmas.*

Honors and Awards

- **ACM Gordon Bell Prize, 2025**
- Peter O'Donnell, Jr. Postdoctoral Fellowship, 2021
- **ACM-IEEE CS George Michael Memorial HPC Fellowship, 2019**
- The University of Utah Graduate Research Fellowship, 2019

Research Interests

- Massively parallel large-scale computations, parallel algorithms, scientific computing, high performance computing, numerical analysis, scientific machine learning.
- Adaptive discretizations, scalable AMR, Parallel in time methods.
- Boltzmann transport equation, computational plasma physics, computational fluid dynamics, numerical relativity, relativistic Boltzmann transport equation

Submitted Papers

1. **Fernando, M.**, Bochkov, D., Almgren-Bell, J., Oliver, T., Moser, R., Varghese, P., Raja, L. and Biros, G., 2024. A fast solver for the spatially homogeneous electron Boltzmann equation. arXiv preprint arXiv:2409.00207 (submitted to the Journal of Computational Physics).

2. **Fernando, M.**, Almgren-Bell, J., Oliver, T., Moser, R., Varghese, P., Raja, L. and Biros, G., 2025. Boltzsim: A fast solver for the 1D-space electron Boltzmann equation with applications to radio-frequency glow discharge plasmas. arXiv preprint arXiv:2502.16555 (submitted to the Journal of Computational Physics).

3. **Fernando, M.**, Oliver, T., Moser, R., Varghese, P., Raja, L. and Biros, G., 2024. A batched electron-Boltzmann solver for low-temperature plasmas (submitted to Computer Physics Communications).

Publications

1. Henneking, S., Venkat, S., Dobrev, V., Camier, J., Kolev, T., **Fernando, M.**, Gabriel, A.A. and Ghattas, O., 2025, November. Real-time Bayesian inference at extreme scale: A digital twin for tsunami early warning applied to the Cascadia subduction zone. In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (**SC'25**)(pp. 60-71).

2. Venkat, S., **Fernando, M.**, Henneking, S. and Ghattas, O., 2025. Fast and scalable FFT-based GPU-accelerated algorithms for block-triangular Toeplitz matrices with application to linear inverse problems governed by autonomous dynamical systems. SIAM Journal on Scientific Computing, 47(5), pp.B1201-B1226.

3. Black, W.K., Neilsen, D., Hirschmann, E.W., Van Komen, D.F. and **Fernando, M.**, 2025. Nyquist-resolving gravitational waves via orbital frequency-based refinement. Physical Review D, 111(12), p.124001.

4. Ruys, W., Lee, H., You, B., Talati, S., Park, J., Almgren-Bell, J., Yan, Y., **Fernando, M.**, Biros, G., Erez, M. and Burtscher, M., 2024, May. A Deep Dive into Task-Based Parallelism in Python. In 2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW) (pp. 1147-1149). IEEE.

5. **Fernando, M.**, Neilsen, D., Zlochower, Y., Hirschmann, E.W. and Sundar, H., 2023. Massively parallel simulations of binary black holes with adaptive wavelet multiresolution. Physical Review D, 107(6), p.064035.

6. **Fernando, M.**, Neilsen, D., Hirschmann, E., Zlochower, Y., Sundar, H., Ghattas, O. and Biros, G., 2022, November. A GPU-accelerated AMR solver for gravitational wave propagation. In SC22: International Conference for High Performance Computing, Networking, Storage and Analysis (**SC'22**) (pp. 1-15). IEEE.

7. Khanwale, M.A., Saurabh, K., **Fernando, M.**, Calo, V.M., Sundar, H., Rossmanith, J.A. and Ganapathysubramanian, B., 2022. A fully-coupled framework for solving Cahn-Hilliard Navier-Stokes equations: Second-order, energy-stable numerical methods on adaptive octree based meshes. Computer Physics Communications, 280, p.108501.

8. Xu, S., Zhu, Q., **Fernando, M.** and Sundar, H., 2022. A finite element level set method based on adaptive octree meshes for thermal free-surface flows. International Journal for Numerical Methods in Engineering, 123(22), pp.5500-5516.

9. Tran, H.D., **Fernando, M.**, Saurabh, K., Ganapathysubramanian, B., Kirby, R.M. and Sundar, H., 2022, May. A scalable adaptive-matrix spmv for heterogeneous architectures. In 2022 IEEE International Parallel and Distributed Processing Symposium (**IPDPS'22**) (pp. 13-24). IEEE.

10. **Fernando, M.**. and Sundar, H., 2022. Scalable Local Timestepping on Octree Grids. SIAM Journal on Scientific Computing, 44(2), (**SIAM SISC**), pp.C156-C183.

11. Saurabh, K., Ishii, M., **Fernando, M.**, Gao, B., Tan, K., Hsu, M.C., Krishnamurthy, A., Sundar, H. and Ganapathysubramanian, B., 2021, November. Scalable adaptive PDE solvers in arbitrary domains. In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (**SC'21**) (pp. 1-15)

12. Saurabh, K., Gao, B., **Fernando, M.**, Xu, S., Khanwale, M.A., Khara, B., Hsu, M.C., Krishnamurthy, A., Sundar, H. and Ganapathysubramanian, B., 2021. Industrial scale Large Eddy Simulations with adaptive octree meshes using immersogeometric analysis. Computers & Mathematics with Applications, 97, pp.28-44.

13. Xu, S., Gao, B., Lofquist, A., **Fernando, M.**, Hsu, M.C., Sundar, H. and Ganapathysubramanian, B., 2021. An octree-based immersogeometric approach for modeling inertial migration of particles in channels. Computers & Fluids, 214, p.104764.

14. Ishii, M., **Fernando, M.**, Saurabh, K., Khara, B., Ganapathysubramanian, B. and Sundar, H., 2019, November. Solving PDEs in space-time: 4D tree-based adaptivity, mesh-free and matrix-free approaches. In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (**SC'19**) (pp. 1-61).

15. **Fernando, M**., Neilsen, D., Lim, H., Hirschmann, E. and Sundar, H., 2018. Massively Parallel Simulations of Binary Black Hole Intermediate-Mass-Ratio Inspirals,SIAM Journal on Scientific Computing, 41(2), (**SIAM SISC**), pp.C97-C138.

16. **Fernando, M.**, Neilsen, D., Hirschmann, E.W. and Sundar, H., 2019, June. A scalable framework for adaptive computational general relativity on heterogeneous clusters. In Proceedings of the ACM International Conference on Supercomputing (**ICS'19**) (pp. 1-12).
17. **Fernando, M.**, Duplyakin, D. and Sundar, H., 2017, June. Machine and application aware partitioning for adaptive mesh refinement applications. In Proceedings of the 26th International Symposium on High-Performance Parallel and Distributed Computing (**HPDPS'19**) (pp. 231-242). ACM.
18. Fernando, I.D., Jayasena, S., **Fernando, M.** and Sundar, H., 2017, August. A Scalable Hierarchical Semi-Separable Library for Heterogeneous Clusters. In 2017 46th International Conference on Parallel Processing (**ICPP'17**) (pp. 513-522). IEEE.

Conference and Seminar Presentations

- Jun. 2025 - North American Einstein Toolkit Workshop 2025, Austin, TX.
- Jun. 2024 - North American Einstein Toolkit Workshop 2024, Baton Rouge, LA.
- Apr. 2024 - 18th Copper Mountain Conference On Iterative and Multigrid Methods, Copper Mountain, CO.
- Mar. 2024 - SIAM Conference on Parallel Processing for Scientific Computing (PP24), Baltimore, MD.
- Dec. 2023 - Numerical Relativity group seminar, Rochester Institute of Technology (RIT) (Virtual).
- Dec. 2022 - Babuška Forum, Oden Institute, Austin, TX.
- Nov. 2022 - The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC'22), Dallas, TX.
- Jun. 2019 - ACM International Conference on Supercomputing, Phoenix, AZ.
- Jun. 2018 - 2018 North American Einstein Toolkit Workshop, Atlanta, GA.
- Jun. 2017 - ACM Symposium on High-Performance Parallel and Distributed Computing, Washington D.C.

Academic Service

- **Activities organized**
  - Mini-symposium on "Kinetic Equation Solvers for Plasmas", SIAM Conference on Parallel Processing for Scientific Computing (PP24), (with James Almgren-Bell, George Biros).
  - Mini-symposium on ""Digital Twins for Large-Scale Complex Systems" SIAM Conference on Computational Science and Engineering (CSE25), (organized with Omar Ghattas, Stefan Henneking, Sreeram Venkat).
  - Mini-symposium on "Digital Twins for Large-Scale Complex Systems", SIAM Conference on Parallel Processing for Scientific Computing (PP24), (organized with Omar Ghattas, Stefan Henneking, Sreeram Venkat).
- **Reviewing**
  - IEEE International Parallel & Distributed Processing Symposium
  - The International Conference for High Performance Computing, Networking, Storage, and Analysis
  - Journal of Parallel and Distributed Computing
  - The Journal of Supercomputing

Teaching and Mentoring Experiences

- I have worked closely with two Ph.D. students at Oden Institute,
  - To develop a particle-in-cell direct simulation Monte-Carlo (PIC-DSMC) method for solving electron Boltzmann transport equation for low-temperature plasmas.

- Designing a parallel algorithm that exploit Block-Toeplitz structure of the operator to enable fast Hessian evaluations.

- Spring 2017, **Teaching Assistant**, Parallel and High Performance Computing, The University of Utah.
- Fall 2015, **Teaching Assistant**, Big Data Computer Systems, The University of Utah.

References

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